

Foundation (Jalapeno)

Open Ended Questions (FRQ)

Q9. A rocket is launched from the ground and its height above the ground is given by the quadratic equation $y = x^2 + 4x + 4$, where y is the height in meters and x is the time in seconds. Write this equation in vertex form and explain what the vertex represents in this context.

Q10. The orbital path of a satellite around the Earth can be modeled by the quadratic equation $y = (x - 2)(x + 4)$, where y is the distance from the

Chef's Tips

- Tip 1: Make sure to check the units of measurement in the word problems.
- Tip 2: Use graphing calculators or software to visualize the parabolas and verify the answers.
- Tip 3: Emphasize the importance of understanding the parameters in vertex and intercept forms.